Year 12 Maths - Statistical hypothesis testing

Similar questions - Two-tailed tests and critical regions

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A spinner has a green sector. The manufacturer says that the probability of the spinner landing on green is 0.3. Ali suspects that this is not correct. He spins the spinner 30 times and it lands on green 4 times.

(a) Test, at the 10% significance level, whether the probability is different from 0.3. **[6]**

(b) Find the critical region for this test. **[3]**

(c) Find the actual significance level of the test. **[1]**

2 Jo wants to test whether a six-sided die is biased. She throws it 60 times, recording the number of sixes, X. She uses a 5% significance level.

(a) State suitable hypotheses, and explain why the test is two-tailed. **[2]**

(b) Find the critical region for the test, using a probability of at most 2.5% in each tail. **[4]**

(c) In Jo's experiment, 15 sixes occur. State the conclusion of the test. **[1]**

(d) Find the probability that the test concludes the die is biased when in fact it is fair. **[1]**

Question 1

(a) $H_0 : p = 0.3$, $H_1 : p \neq 0.3$. $X \sim B(30, 0.3)$. Two-tailed at 10%: 5% in each tail. Observed 4 is in the lower tail: $P(X \leqslant 4) = 0.0302 < 0.05$. Reject H_0: there is evidence that the probability of green is different from 0.3

(b) Lower tail: $P(X \leqslant 5) = 0.0766 > 0.05$, $P(X \leqslant 4) = 0.0302$, so $X \leqslant 4$. Upper tail: $P(X \geqslant 13) = 0.0845 > 0.05$, $P(X \geqslant 14) = 0.0401$, so $X \geqslant 14$. Critical region: $X \leqslant 4$ or $X \geqslant 14$

(c) $0.0302 + 0.0401 = 0.0703$, i.e. 7.03%

Question 2

(a) $H_0 : p = \frac{1}{6}$, $H_1 : p \neq \frac{1}{6}$. The die could be biased towards or against sixes, so there are two tails

(b) $X \sim B\left(60, \frac{1}{6}\right)$. Lower tail: $P(X \leqslant 4) = 0.0202 < 0.025$ but $P(X \leqslant 5) = 0.0512 > 0.025$. Upper tail: $P(X \geqslant 17) = 0.0164 < 0.025$ but $P(X \geqslant 16) = 0.0338 > 0.025$. Critical region $X \leqslant 4$ or $X \geqslant 17$

(c) 15 is not in the critical region: do not reject H_0, so there is no evidence at the 5% level that the die is biased

(d) $0.0202 + 0.0164 = 0.0366$ (the actual significance level, 3.66%)